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United States Patent Application Publication	20250282510
Kind Code	A1
Publication Date	September 11, 2025
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MACHINE FOR WRAPPING A LOAD

Abstract

A wrapping machine for wrapping a load with a paper band has a table to support the load and rotatable about a wrapping axis, an unwinding apparatus including a supporting frame to rotatably support a reel of paper band and rollers to unwind and guide the paper band, a moving assembly including a moving carriage movable parallel to the wrapping axis and supporting the unwinding apparatus rotatably about an inclination axis, and a pressing and cutting system including a pressing assembly and a cutting assembly respectively for pressing a portion of paper band on the load at the end of the wrapping cycle and for cutting the paper band. The pressing assembly includes pressing rollers adjacent to each other and parallel to the wrapping axis and the cutting assembly includes a blade mounted on a driving carriage linearly movable along a cutting direction.

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Family ID:	**82608258**
Appl. No.:	**18/860815**
Filed (or PCT Filed):	**April 26, 2023**
PCT No.:	**PCT/IB2023/054311**

Foreign Application Priority Data

IT	102022000008714	Apr. 29, 2022

Publication Classification

Int. Cl.: **B65B11/04** (20060101)

U.S. Cl.:

CPC **B65B11/045** (20130101);

Background/Summary

[0001] The invention relates to wrapping machines arranged for wrapping a load with a paper band.
[0002] The use of paper sheets and bands for packaging products, typically envelopes and bags, is known in the packaging sector. In particular, the use of Kraft-type paper, also called wrapping paper, for packaging is widely used, since this paper is characterized by a high resistance, in particular to traction and punctures and tears, and longitudinal and transverse elasticity. Kraft paper can be smooth or have creases or micro-creases that accentuate the elasticity thereof.
[0003] By virtue of its mechanical features and its complete recyclability, Kraft paper can also be validly used for wrapping palletized loads, i.e., loads formed by a single product or by a plurality of products grouped and arranged on a pallet, in place of the films of cold-extensible plastic material generally used. In fact, the need to reduce the use and consumption of plastics is increasingly felt also in the field of wrapping palletized products, which, as is known, if not correctly handled and above all disposed of, significantly contribute to environmental pollution. Paper is instead an eco-friendly and eco-sustainable material obtainable both from recycling pre-existing paper and from sustainable productions.
[0004] The known wrapping machines for wrapping with paper typically comprise an unwinding apparatus which is movable parallel and/or around a wrapping axis and which supports a reel of paper band from which the paper band is unwound to be wrapped around the load so as to form a series of strips or bands, generally having a helical trend, by virtue of the combination of the relative linear and rotational movements between the unwinding apparatus and the load.
[0005] In wrapping machines provided with a rotating table to support the load, during the wrapping cycle the latter is rotated around a vertical wrapping axis, while the unwinding apparatus is moved parallel to the wrapping axis with alternating motion along a fixed column of the wrapping machine.
[0006] In wrapping machines with horizontal rotating ring or rotating arm, the load remains static during the wrapping, while the unwinding apparatus is moved with respect to the latter both in rotation around a vertical winding axis and in translation parallel to the latter. To this end, the unwinding apparatus is fixed to a ring structure or to an arm rotatably supported by a machine frame and so as to rotate around the load.
[0007] In wrapping machines with a vertical rotating ring the load is moved horizontally through the ring structure, while the unwinding apparatus fixed to the ring structure is rotated around a horizontal axis.
[0008] The unwinding apparatus is generally provided with a series of rollers for diverting the unwound paper band from the reel towards the load during the unwinding.
[0009] In order to avoid folding and creasing in the paper during the wrapping, the unwinding apparatus is generally mounted on the wrapping machine so that it can be inclined about a horizontal axis by an angle substantially equal to the helical wrapping angle, such a helical angle being a function, as known, of the vertical linear displacement of the unwinding apparatus in a wrapping revolution.
[0010] To this end, different solutions are known for tilting the unwinding apparatus, which are generally quite complex and expensive, as they comprise motorizations or mechanical systems with

elastic returns.
[0011] At the end of the wrapping the paper band must be fixed to the load and then cut to obtain a first paper flap or end fixed to the load and a second paper end connected to the reel and retained by special gripping means.
[0012] In particular, the first paper end is positioned by suitable pressing means on the load and fixed to the layers of paper already wrapped around the load by means of glue or adhesive.
[0013] Cutting means are also included for cutting the paper band.
[0014] A problem of the pressing and cutting means of the known wrapping machines lies in the difficulty of executing a precise and correct cut of the paper and in making the first paper end, which is obtained from the cut, optimally adhere to the load.
[0015] EP 3835220 discloses a machine for wrapping one or more products with a band of material comprising a support on which a main reel is rotatably positioned, from which the material band is unwound, a vertical guide and a wrapping head movable along the vertical guide. The wrapping head comprises at least one wrapping/unwinding roller rotatable in one direction to allow the wrapping therearound of a predetermined length of the material band and rotatable in the opposite direction to be able to unwind at least in part the band wrapped therearound so that a prefixed length of band can be unwound from the main reel to be wrapped around the wrapping/unwinding roller to execute a wrapping by subsequently unwinding at least one part of the aforesaid band.
[0016] An object of the invention is to improve the known wrapping machines arranged to wrap a load with a paper band.
[0017] Another object is to provide a wrapping machine which allow any type of load to be wrapped firmly and compactly by means of a paper band, avoiding the formation of folds and creases on said band during the wrapping and reducing and almost eliminating the risks of its breakage or tearing.
[0018] A further object is to provide a wrapping machine which allow to optimally fix the various tapes or strips of paper band wrapped around the load and in particular an end of such a paper band so as to achieve a firm, compact and robust wrapping of the load, in particular of a palletized load formed by a plurality of products grouped and overlapped on a pallet.
[0019] A still further object is to provide a wrapping machine which allow to precisely and accurately cut the paper band at the end of the wrapping and to optimally fix an end of the paper band obtained from cutting to a side wall of the load.
[0020] Another further object is to provide a wrapping machine having simple and economical construction and precise and reliable operation.
[0021] These objects and others are achieved by a wrapping machine according to claim **1**.

Description

[0022] The invention can be better understood and implemented with reference to the attached drawings which illustrate exemplifying and non-limiting embodiments thereof, in which:
[0023] FIG. **1** is a perspective view of the wrapping machine of the invention in particular provided with an unwinding apparatus of a paper band;
[0024] FIG. **2** is an enlarged perspective side view of the wrapping machine of FIG. **1** illustrating in particular the unwinding apparatus;
[0025] FIG. **3** is a partial side view of the wrapping machine of FIG. **1**;
[0026] FIG. **4** is an enlarged perspective view of the unwinding apparatus;
[0027] FIG. **5** is a perspective view from a different angle of the unwinding apparatus of FIG. **4**;
[0028] FIG. **6** is a plan view from above of the unwinding apparatus of FIG. **5**;
[0029] FIGS. **7** and **8** are respectively front and rear perspective views of linking elements of the unwinding apparatus of FIG. **1**;

[0030] FIG. **9** is a rear view of the linking elements of FIGS. **7** and **8**;
[0031] FIG. **10** is a section according to the plane X-X of FIG. **9**;
[0032] FIGS. **11** and **12** are schematic front views illustrating respective steps of a wrapping cycle of a load executed by the wrapping machine of FIG. **1**;
[0033] FIG. **13** is a partial perspective view of the wrapping machine of the invention illustrating in particular a system for pressing and cutting the paper band in an operative configuration;
[0034] FIG. **14** is another partial perspective view from a different angle of the pressing and cutting system of FIG. **13** in the operative configuration and without upper covering means;
[0035] FIG. **15** is an enlarged partial perspective view of the pressing and cutting system of FIG. **13** illustrating upper covering means associated with the cutting means;
[0036] FIG. **16** is a partial plan view from above of the pressing and cutting system of FIG. **13** associated with a paper band and lacking upper covering means;
[0037] FIG. **17** is a partial perspective view of the pressing and cutting system of FIG. **13** lacking upper covering means illustrating in particular the cutting means;
[0038] FIG. **18** is a perspective view of the wrapping machine of the invention associated with a feeding unit of adhesive substance.
[0039] Referring to FIGS. **1** to **17**, the wrapping machine **100** of the invention is arranged to wrap a load **200** with a paper band **50** unwound from a reel **51**. The paper band **50** is for example a band of a known and commercially available Kraft-type paper.
[0040] The wrapping machine **100** comprises a table **120** adapted to support the load **200**, for example formed by a single product or by a plurality of products grouped and arranged on a pallet, and rotatable around a wrapping axis W, in particular substantially vertical. The wrapping machine **100** also comprises an unwinding apparatus **1** and moving means **101** for supporting and moving the unwinding apparatus **1** parallel to the wrapping axis W.
[0041] In particular, the unwinding apparatus **1** comprises a supporting frame **2** arranged to rotatably support the paper reel **51** and roller means **20** for unwinding from the reel **51** and guiding the paper band **50**, in particular towards the load **200** during the wrapping thereof. The moving means **101** comprise a moving carriage **102** movable parallel to the wrapping axis W, in particular along a support column **103** of the wrapping machine **100**, and supporting the wrapping apparatus **1** rotatably around an inclination axis S, in particular substantially orthogonal to the wrapping axis W.
[0042] With particular reference to FIGS. **3** to **10**, the unwinding apparatus **1** comprises a first linking element **3** fixed to the supporting frame **2** and a second linking element **4** fixed to the moving carriage **102**. The first and the second linking element **3**, **4** are rotatably connected to each other about the inclination axis S at respective upper portions **3***a*, **4***a* by means of a linking pin **5**. More precisely, the linking pin **5** is mounted on respective rolling bearings **28** mounted on respective seats made on the upper portions **3***a*, **4***a* of the first and the second linking element **3**, **4**.
[0043] A ball supporting element **6** is interposed between lower portions **3***b*, **4***b* of the linking elements **3**, **4** for slidably supporting, in cooperation with the linking pin **5**, the weight of the unwinding apparatus **1** and facilitating the rotation thereof about the inclination axis S during a wrapping cycle of the load **200**.
[0044] More precisely, the ball supporting element **6** is rotatably housed in a seat **7** fixed, for example, in a respective lower portion **4***b* of the second linking element **4** to abut a rear wall **3***c* of the first linking element **3**.
[0045] Alternatively, the ball supporting element **6** can be rotatably housed in a seat **7** fixed in a respective lower portion **3***b* of the first linking element **3** to abut a rear wall **4***c* of the second linking element **4**.
[0046] The unwinding apparatus **1** further comprises an abutment pin **8**, in particular substantially parallel to the inclination axis S, fixed to a respective rear wall **3***c*, **4***c* of the first linking element **3** or of the second linking element **4** and configured to engage in a slot **9** made in a respective rear

wall **4**_c_, **3**_c_ of the second linking element **4** or of the first linking element **3** to limit an inclination of the unwinding apparatus **1** about the inclination axis S.
[0047] In the embodiment shown in the figures, the abutment pin **8** is fixed to the rear wall **3**_c_ of the first linking element **3** and the slot **9**, in particular through, is made in the rear wall **4**_c_ of the second linking element **4**.
[0048] The slot **8** has an arcuate shape, and in particular it extends along an arc of circumference having a centre substantially coincident with the inclination axis S.
[0049] The slot **8** has longitudinal dimensions or lengths such as to allow the unwinding apparatus **1** to be rotatable about the inclination axis S, with respect to a reference plane H passing through the inclination axis S and the abutment pin **8** and with respect to the wrapping axis W, in both directions of rotation (counter-clockwise and clockwise with reference to FIGS. **11** and **12**) at the most respectively by a first angle of maximum inclination α**1** and by a second angle of maximum inclination α**2**. The first angle of maximum inclination α**1** corresponds to the angle of maximum inclination of the unwinding apparatus **1** during an upwards movement thereof, in one step of the wrapping cycle of the load **100**, while the second angle of maximum inclination α**2** corresponds to the angle of maximum inclination of the unwinding apparatus **1** during a downwards movement thereof in another step of the wrapping cycle of the load **100**.
[0050] As better explained in the following description, the unwinding apparatus **1** is inclined by the same paper band **50** during the wrapping of the load **200** so that the paper band is also inclined (more precisely its longitudinal edges **50**_c_ are inclined) with respect to the wrapping axis W by an angle (at the most equal to the first angle of maximum inclination α**1** or the second angle of maximum inclination α**2**) substantially equal to a wrapping angle β of the paper band **50** around the load **200**, so as to avoid the formation of folds and/or creases on the paper band **50** itself.
[0051] As known, the wrapping angle β, or helical wrapping angle, is the angle formed by a longitudinal edge **50**_c_ of the paper band **50** with a reference plane M orthogonal to the wrapping axis W, i.e., substantially horizontal, said wrapping angle β being a function of a vertical linear displacement of the unwinding apparatus **1** in a wrapping revolution.
[0052] The angles of maximum inclination α**1**, α**2** are comprised between 2° and 10° (sexagesimal degrees). The first angle of maximum inclination α**1** and the second angle of maximum inclination α**2** can be the same or different, for example the second angle of maximum inclination α**2** can be less than the first angle of maximum inclination α**1**.
[0053] Each linking element **3**, **4** comprises a respective elongated metal profile having a "U"-shaped transverse section and formed by a respective bottom wall **13**, **43** and two side walls **14**, **44**. The two bottom walls **13**, **43** of the linking elements **3**, **4** are facing and the linking pin **5**, the abutment pin **8** are fixed thereon, the slot **9** is made and the ball supporting element **6** is fixed.
[0054] The roller means **20** of the unwinding apparatus **1** comprise a plurality of guiding rollers **21** and an unwinding roller **22** controllable in rotation speed. In particular, the unwinding roller **2**) is associated with a brake or a motor, so as to control a traction or "pulling" force of the paper band **50** during the wrapping cycle, in particular so as to keep the value of said traction force within a predefined range in order to avoid, on the one hand, the breakage of the paper band, on the other hand, a loose wrapping of the load.
[0055] With particular reference to FIGS. **2** and **26**, the unwinding apparatus **1** further comprises at least one dispensing unit **15** of an adhesive substance adjustably fixed to a supporting rod **26** adjacent to a guiding roller **21** of the roller means **20** on which the paper band **50** exiting the unwinding apparatus **1** is wrapped.
[0056] In the illustrated embodiment, the unwinding apparatus **1** is provided with a pair of dispensing units **15**, each of which is arranged to dispense dots or lines of fixed length of adhesive substance on portions of the paper band **50** intended to be wrapped on edges **200**_a_ of the load **200**. The adhesive substance is, for example, a cold glue, provided to the dispensing units **15**, for example, by an external feeding unit **60** which is separate and independent from the wrapping

machine, as illustrated in FIG. **18**.

[0057] The feeding unit **60** for feeding the dispensing units **15** with glue comprises a movable carriage **61** on wheels which houses a glue tank, a pump for withdrawing from the tank and sending the glue under pressure to the dispensing units **15** and a local control unit, for example a respective industrial computer or PLC, connected to the control unit of the wrapping machine **100** to control the operation of the pump, in particular to feed the dispensing units **15** with glue when required.

[0058] Alternatively, the feeding unit **60** can be mounted directly on the wrapping machine **100**, for example associated with the unwinding unit **1**.

[0059] With particular reference to FIGS. **13** to **17**, the wrapping machine **100** of the invention also comprises a pressing and cutting system **10** of the paper band **50** which comprises pressing means **11** and cutting means **12** configured respectively to press a portion of the paper band **50** on the load **200**, more precisely on a side wall thereof, at the end of the wrapping cycle and then cut said paper band **50** so as to make a first paper flap or end which is fixed to the load **200** (to its side wall) and a second paper end which is connected to the reel **50**, in particular it is held by suitable gripping means **130** of the wrapping machine **100**, of known type and not described in detail, placed adjacent to the rotating table **120**.

[0060] The pressing means **11** comprise at least one pair of pressing rollers **31** mounted idle, i.e., free to rotate about respective longitudinal axes, on a mounting frame **38** of the pressing and the cutting system **10**, adjacent to each other and parallel to the wrapping axis W.

[0061] The cutting means **12** comprise at least one blade **32** mounted on a driving carriage **33** which is movable linearly, in particular parallel to the wrapping axis W, so that in operation the blade **32** can abut and cut the paper band **50** along a cutting direction T transversal, in particular substantially orthogonal, to the external longitudinal edges **50***c* of the paper band.

[0062] The cutting means **12** also comprise a guiding element **34** provided with a abutment wall **34***a* that is substantially flat and parallel to the wrapping axis W and configured to engage and guide the paper band **50** coming from the unwinding apparatus **1** and moving towards the pressing means **11**. The blade **32** is adjacent to the guiding element **34** and arranged and movable along a cutting plane T**1**, which is substantially parallel to the wrapping axis W, i.e., substantially vertical, and inclined with respect to the abutment wall **34***a* of the guiding element **34** by a cutting angle γ comprised between 70° and 90° (FIG. **16**).

[0063] As better explained in the following description, by virtue of the guiding element **34** and the particular arrangement of the blade **32**, the paper band **50**, suitably stretched between the cutting means **12** and the unwinding apparatus **1**, can be cut quickly, cleanly and precisely. The blade **32** is reversibly fixed to the driving carriage **33** so as to be easily and quickly replaceable when worn and/or damaged.

[0064] The cutting means **12** comprise driving means **35** arranged to linearly move the driving carriage **33** along the cutting direction T. In the illustrated embodiment, the driving means **35** comprise a rotary electric motor **36** adapted to drive screw-nut means **39**, of a known type and not described in detail, in rotation via motion transmission means **37**, associated with and operating on the driving carriage **33**.

[0065] The pressing and cutting system **10** comprises a first supporting arm **16** rotatably mounted about a first rotation axis X**1**, in particular on a base **140** of the wrapping machine **100**, so as to rotate the entire pressing and cutting system **10** in an operative position A, in which the pressing means **11** and the cutting means **12** are closer to the load **200**. The first supporting arm **16** is also arranged to rotate the pressing and cutting system **10** in an inoperative position, not illustrated in the figures, in which the pressing means **11** and the cutting means **12** are spaced from the load **100** so as not to interfere with its wrapping.

[0066] The pressing and cutting system **10** also comprises a second supporting arm **17** slidably supported by the first arm **16** and having a free end **17***a* supporting the pressing means **11** and the cutting means **12**. In particular, the second supporting arm **17** is linearly movable to allow the pressing means **11** to abut the load **200**, when the first supporting arm **16** is rotated in the operative position A.

[0067] The pressing and cutting system **10** further comprises covering means **40** arranged to contain and enclose the driving means **35** and provided with a housing **41** capable of housing and enclosing the blade **32** arranged in an inoperative position B. Thus, the blade **32** is inaccessible to an operator for his safety and security.

[0068] Sensor means of known type and not illustrated in the figures are associated with the cutting means **12** and are configured to detect at least the inoperative position B of the blade **32**, and preferably a start and end stroke position of the driving carriage **33**.

[0069] The operation of the wrapping machine **100** of the invention includes during a wrapping cycle of the load **200** that the unwinding apparatus **1** delivers the paper band **50** unwound from the reel **50** by virtue of the rotation about the wrapping axis W of the rotating table **120**, on which the load **200** is positioned.

[0070] An initial end of the paper band **50** is previously fixed to the load **200** by virtue of the gripping means **130** which arranged in a raised position retain an initial end of paper band (for example obtained by cutting the paper band in a previous wrapping cycle) so that it is covered and pressed by one or more bands of paper band wrapped around the load, according to a well-known mode used in rotating table wrapping machines.

[0071] During the wrapping, the unwinding apparatus **1** is moved by the moving means **101** with reciprocating motion parallel to the rotation axis W so that the load **200** is wrapped by a plurality of partially overlapping paper strips or bands having a helical trend, by virtue of the combination of the relative, linear and rotation movements, between the unwinding apparatus and the load. In particular, the paper band **50** is wrapped with a predefined wrapping angle as a β (function of the vertical linear displacement of the unwinding apparatus **1** in a wrapping revolution).

[0072] Since the unwinding apparatus **1** is mounted on the moving carriage **102** of the moving means **101** in a rotatable manner about the inclination axis S, during the wrapping it can be inclined by the same paper band being wrapped so that the latter is inclined (more precisely its longitudinal edges **50***c* are inclined) with respect to the wrapping axis W by an inclination angle α which is substantially equal to the wrapping angle β so as to avoid the formation of folds and/or creases on the paper band **50** itself.

[0073] In particular, by virtue of the first linking element **3** and the second linking element **4** (fixed respectively to the supporting frame **2** of the unwinding apparatus **1** and to the moving carriage **102**) connected to each other in a rotatable manner about the inclination axis S by the linking pin **5** and by interposing the ball supporting element **6**, the unwinding apparatus **1** can be inclined in a smooth and stable manner during the wrapping. It should be noted that the ball supporting element **6**, interposed between the lower portions **3***b*, **4***b* of the linking elements **3**, **4**, in addition to allowing to slidably support, in cooperation with the linking pin **5**, the weight of the unwinding apparatus **1**, allows the latter to rotate precisely and accurately, avoiding in particular linear movements parallel to the inclination axis S which could compromise the correct dispensing of the paper band.

[0074] During the wrapping, the dispensing units **15**, fed by the feeding unit **60**, intermittently dispense adhesive substance onto the paper band **50** exiting from the unwinding apparatus **1**. More precisely, dispensing units **15** are activated and controlled so as to deliver dots or lines of fixed length of an adhesive substance on the portions of the paper band **50** intended to be wrapped on the edges **200***a* (i.e., the vertical lines defined by the junction of two respective side walls) of the load **200**.

[0075] Numerous tests carried out by the applicant have shown that when the overlapping strips or tapes of paper band **50** adhere and are joined together mainly at the edges **200***a* of the load **200**, the resulting wrapping is more stable, robust and firm.

[0076] Upon completion of the wrapping of the load **200** with a defined number of strips or tapes of paper band **50**, the rotating table **120** is stopped and the unwinding apparatus **1** is arranged in a

lowered position. The pressing and cutting system **10** is activated so as to press with the pressing rollers **31** of the pressing means **11** the portion of paper band coming from the wrapping apparatus **1** against a side wall of the load **100**.
[0077] More precisely, the first supporting arm **16** of the pressing and cutting system **10** is rotated about the first rotation axis X**1** in the operative position A and the second supporting arm **17** is linearly moved towards the load **200** to push the paper band by means of the pressing rollers **31** against the side wall of the load **200**. The portion of paper band which is pushed against the load **200** was previously sprinkled of adhesive substance by dispensing units (exiting from the unwinding apparatus). The pressing means **11** maintain the pressure for a defined time interval (for example from 4 to 10 s) in order to allow the adhesive substance to grip and firmly join the aforesaid portion of paper band with the underlying paper strips. At the same time, the cutting means **12** are activated to cut the paper band and make a first paper end which is fixed to the load **200** (to its side wall) and a second paper end which is connected to the reel **50**, in particular it is retained by the gripping means **130** of the wrapping machine **100** activated in a raised position so as to grip and retain the paper band.
[0078] More precisely, the blade **32** of the cutting means **12** mounted on the driving carriage **33** is moved from the inoperative position B along the cutting direction T, substantially orthogonal to the outer longitudinal edges **50***c* of the paper band, in particular with a downwards movement, so as to intercept and cut the paper band **50**. The paper band abuts and engages the abutment wall **34***a*, which is substantially flat and parallel to the wrapping axis W, of the guiding element **34**, with respect to which the blade **32** is inclined by a cutting angle γ comprised between 70° and 90°. Therefore, by virtue of the guiding element **34** and the particular arrangement of the blade **32**, the paper band **50**, suitably stretched between the cutting means **12** and the unwinding apparatus **1**, can be cut quickly, cleanly and precisely. It should be noted that in the inoperative position B the blade **32** is housed inside the housing **41** of the covering means **40** so as not to be accessible to an operator.
[0079] Further described is an unwinding apparatus **1** associable with a wrapping machine **100** for dispensing a paper band **50** with which to wrap a load **200**.
[0080] The unwinding apparatus **1**, illustrated in particular in FIGS. **2** to **10** and already described above, comprises a supporting frame **2** arranged to rotatably support a reel **51** of the paper band **50** and roller means **20** for unwinding the reel **51** and guiding the paper band **50**. The unwinding apparatus **1** comprises in particular a first linking element **3** fixed to the supporting frame **2** and a second linking element **4** which can be fixed to moving means **101** of the wrapping machine **100** so as to be movable at least parallel to a wrapping axis W, in particular substantially vertical.
[0081] The first and the second linking element **3**, **4** are connected to each other in a rotatable manner about an inclination axis S, in particular substantially orthogonal to the wrapping axis W, at respective upper portions **3***a*, **4***a* by means of a linking pin **5**. A ball supporting element **6** is interposed between the lower portions **3***b*, **4***b* of the linking elements **3**, **4** for slidably supporting in cooperation with the linking pin **5** the weight of said unwinding apparatus **1** and facilitating the rotation thereof about the inclination axis S during operation. In particular, the ball supporting element **6** is rotatably housed in a seat **7** e.g., fixed to a respective lower portion **4***b* of the second linking element **4** and is configured to abut a rear wall **3***c* of the first linking element **3**.
[0082] The unwinding apparatus **1** further comprises an abutment pin **8** fixed to a respective rear wall **3***c*, **4***c* of the first linking element **3** or of the second linking element **4** and configured to engage in a slot **9**, in particular through, made in a respective rear wall **4***c*, **3***c* of the second linking element **4** or of the first linking element **3** to limit an inclination of the unwinding apparatus **1** about the inclination axis S. The abutment pin **8** is substantially parallel to the inclination axis S.
[0083] Thereby, the unwinding apparatus **1** is rotatable about the inclination axis S, with respect to a reference plane H passing through the inclination axis S and the abutment pin **8** in the two rotation directions at the most respectively by a first angle of maximum inclination α**1** and by a

second angle of maximum inclination α**2**.
[0084] The aforesaid angles of maximum inclination α**1**, α**2** are comprised between 2 and 10.
[0085] Each linking element **3**, **4** comprises a respective elongated metal profile having a “U”-shaped transverse section and formed by a respective bottom wall **13**, **43** and two side walls **14**, **44**. The two bottom walls **13**, **43** of the linking elements **3**, **4** are facing and the linking pin **5**, the abutment pin **8** are fixed thereon, the slot **9** is made and the ball supporting element **6** is fixed.
[0086] The unwinding apparatus **1** can be mounted on a wrapping machine of the rotating table type, in particular fixed to a moving carriage of the moving means **101** of the wrapping machine, movable parallel to the wrapping axis W, in particular along a support column.
[0087] Alternatively, the unwinding apparatus **1** can be mounted on a wrapping machine with horizontal rotating ring or rotating arm in which the load remains static during the wrapping, while the unwinding apparatus is moved with respect to the latter both in rotation around a vertical wrapping axis and in translation along the latter. In particular, the unwinding apparatus **1** can be fixed to a ring structure or to an arm rotatably supported by a machine frame and so as to rotate around the load.
[0088] The unwinding apparatus **1** can also be mounted on a vertical wrapping machine with rotating ring in which the load is moved horizontally through the ring structure, while the unwinding apparatus fixed to the latter is rotated about a horizontal axis.
[0089] Lastly, the unwinding apparatus **1** can be mounted on a self-propelled wrapping machine, or a self-propelled wrapping robot, in particular slidingly connected to a vertical upright fixed to a self-propelled carriage or cart provided with motorized traction wheels and a guiding device comprising one or more steerable wheels, operated by a steering wheel.
[0090] Further described is a method for wrapping a load **200** with a paper band **50** by means of the wrapping machine **1** described above and illustrated in FIGS. **1** to **17**. Such a method comprises the following steps: [0091] wrapping the paper band **50** unwound from the reel **51** around the load **200** by rotating one with respect to the other about a wrapping axis W, the load **200** and an unwinding apparatus **1** supporting the reel **51** of the wrapping machine **100**; [0092] adjusting during said wrapping an inclination of the paper band **50** with respect to the wrapping axis W by tilting the unwinding apparatus **1** about an inclination axis S by an angle substantially equal to a wrapping angle β of the paper band **50** about the load **200** so as to avoid bending and/or creasing on the paper band **50**; [0093] delivering dots or lines of fixed length of an adhesive substance on portions of the paper band **50** intended to be wrapped on edges **200***a* of the load **200**.
[0094] In particular, the adjusting by tilting said unwinding apparatus **1** is achieved by the paper band **50** being wrapped around the load **200**, the unwinding apparatus **1** being free to rotate about the inclination axis S.
[0095] The method further comprises adjusting a traction or “pulling” force exerted by the paper band **50** on the load **200** during the wrapping of the latter with the paper band **50**, in particular by adjusting a rotation speed of an unwinding roller **22** of roller means **20** of the unwinding apparatus **1**, so as to avoid breakage of the paper band **50** (in case of excessive traction or tension force) or a loose wrapping of the load (in case of insufficient traction or tension force).

Claims

1-10. (canceled)
11. A wrapping machine for wrapping a load with a paper band unwound from a reel comprising: a table adapted to support said load and rotatable about a wrapping axis; an unwinding apparatus comprising a supporting frame arranged to rotatably support said reel of paper band and rollers suitable for unwinding from said reel and guiding said paper band; a moving assembly comprising a moving carriage movable parallel to said wrapping axis and supporting said unwinding apparatus rotatably about an inclination axis; a pressing and cutting system comprising a pressing assembly

and a cutting assembly respectively for pressing a portion of said paper band on the load at the end of the wrapping cycle and for cutting said paper band so as to make a first paper end fixed to said load and a second paper end connected to the reel; wherein said pressing assembly comprises a pair of pressing rollers mounted idle on a mounting frame of said pressing and cutting system, adjacent to each other and parallel to said wrapping axis, and said cutting assembly comprises at least one blade mounted on a driving carriage linearly movable, said blade being configured for abutting and cutting said paper band along a cutting direction that is transverse to external longitudinal edges of said paper band.

12. The wrapping machine according to claim 11, wherein said cutting assembly further comprises a guiding element provided with an abutment wall that is substantially flat and parallel to said wrapping axis and configured to engage and guide towards said pressing assembly said paper band paper exiting the unwinding apparatus, said blade being positioned adjacent to said guiding element and arranged and mobile along a cutting plane that is substantially parallel to said wrapping axis and inclined with respect to said abutment wall of said guiding element by a cutting angle included between 70° and 90°.

13. The wrapping machine according to claim 11, wherein said cutting assembly comprises a driving system for linearly moving said driving carriage along said cutting direction.

14. The wrapping machine according to claim 13, wherein said driving system comprises a rotary electric motor able to rotate by means of a motion transmission assembly a screw-nut arrangement associated with said driving carriage.

15. The wrapping machine according to claim 13, wherein said pressing and cutting system comprises a covering assembly arranged to contain and enclose said driving system and provided with a housing adapted to house and enclose said blade when arranged in an non-operative position.

16. The wrapping machine, according to claim 11, comprising a sensor device associated with said cutting assembly and configured to detect at least one non-operative position of said blade.

17. The wrapping machine according to claim 11, wherein said pressing and cutting system comprises a first supporting arm rotatably mounted about a first rotation axis to rotate said pressing and cutting system in an operating position, in which said pressing assembly and said cutting assembly are closer to said load, and a second supporting arm slidably supported by said first supporting arm, having a free end supporting said pressing assembly and said cutting assembly and movable in an extended position in which said pressing assembly abuts said load.

18. The wrapping machine according to claim 11, wherein said unwinding apparatus comprises a first linking element fixed to said supporting frame and a second linking element fixed to said moving carriage, said first and said second linking elements being connected to each other in a rotatable way about said inclination axis at respective upper portions through a linking pin, said unwinding apparatus further comprising a ball supporting element interposed between lower portions of said linking elements and able to slidably support, in cooperation with said linking pin, a weight of said unwinding apparatus and to facilitate a rotation thereof about said inclination axis during a wrapping cycle of said winding machine.

19. The wrapping machine according to claim 18, wherein said ball supporting element is rotatably housed in a seat fixed to a respective lower portion of said second linking element or of said first linking element and is configured to slidably abut a rear wall of said first linking element or of said second linking element.

20. The wrapping machine according to claim 18, wherein said unwinding apparatus comprises an abutment pin fixed to a respective rear wall of said first linking element or of said second linking element, said abutment pin being substantially parallel to said inclination axis and configured to engage a slot made in a respective rear wall of said second linking element or of said first linking element to limit an inclination of said unwinding apparatus about said inclination axis.

21. The wrapping machine according to claim 11, wherein said second paper end connected to the reel is held by a suitable gripping assembly (**130**) of said wrapping machine.